STAGE 1: PREDICTING POWER OUTPUT OF LARGE-SCALE WAVE ENERGY FARMS USING SPATIAL LAYOUT DATA

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Section 1: Problem

1.1 Description

Renewable and clean energy have become a major trend globally as countries attempt to reduce carbon footprints and transition completely towards clean energy over the years. Various renewable energy sources, including wind, solar, and ocean energy, have been widely studied. Ocean energy, particularly wave energy, has been considered one of the most effective clean energy sources that has attracted increasing attention for sustainable electricity generation.

Australia has significant potential for wave energy generation due to its unique geographical features with a long coastline and strong ocean waves, especially along the southern coast. Government-supported studies, such as the Australian Wave Energy Atlas developed by the Commonwealth Scientific and Industrial Research Organization (CSIRO) and supported by the Australian Renewable Energy Agency (ARENA), have recorded decades of wave data and analyzed substantial wave energy resources across the Australian coastline [1]. Previous research has also indicated that approximately 11% of Australia's electricity generation could potentially be supplied through wave energy by 2050 [2].

However, determining the optimal placement of wave energy converters has been one of the key challenges in developing large-scale wave energy farms. The total power output of a wave energy farm can be greatly influenced by the spatial arrangement of these converters. Previous studies have measured the performance of wave energy converters across the Australian coastal regions, highlighting the importance of location and spatial factors in energy production [3]. Therefore, researching and predicting the total power generation based on the spatial positions of converters is an important research problem for improving the efficiency of wave energy generation.

1.2 Research Questions

1. Can the total power output of a wave energy farm be predicted using the spatial coordinates of wave energy converters?
2. How does the spatial arrangement of wave energy converters influence the overall power generation of the wave farm?
3. Which converter locations contribute most significantly to total power output?

Section 2: Literature review

Optimisation of large wave farms using a multi-strategy evolutionary framework

Neshat et al. (2020) investigated the optimisation of large wave farms consisting of fully submerged three-tether WECs with the aim of maximising total harvested power. The findings demonstrate that the spatial positions of WECs significantly influence the overall power output of wave energy farms, supporting the importance of analysing converter placement in wave energy research. [4]

Layout optimisation of wave energy converter arrays

Mercadé Ruiz et al. (2017) examined the optimisation of WEC array layouts while considering practical engineering constraints such as minimum separation distance between devices and limited deployment areas. Their results show that these constraints must be taken into account when designing feasible wave energy farms, as they influence both installation practicality and overall energy performance. [5]

Wave energy converter array optimization: A genetic algorithm approach and minimum separation distance study.

Sharp and DuPont (2018) further explored WEC array optimisation using a genetic algorithm framework. Their study highlighted the importance of wave interactions, including radiated and diffracted waves between devices, which affect the efficiency of energy capture. The results demonstrated that device spacing and arrangement play a critical role in determining array performance. [6]

Layout study of wave energy converter arrays by artificial neural network and adaptive genetic algorithm.

More recent studies have also investigated data-driven approaches to analysing wave energy systems. Zhu et al. (2022) applied artificial neural networks combined with optimisation algorithms to study the relationship between WEC spatial configurations and power generation. Their research showed that machine learning models can effectively estimate the total energy production of wave farms based on layout information. [7]

Meta Wave Learner: Predicting wave farms power output using effective meta-learner deep gradient boosting model

In addition, Neshat et al. (2024) proposed a machine learning model called the "Meta Wave Learner" to predict wave farm power output using spatial WEC coordinates and energy production data from Australian coastal sites such as Perth and Sydney. Their results demonstrated strong predictive performance compared with other machine learning models, highlighting the potential of data-driven methods in wave energy analysis. [8]

Wave energy converter array layout optimization: A comprehensive review.

Finally, Yang et al. (2022) provided a comprehensive review of WEC array layout optimization studies. The authors analysed numerous research works and concluded that the spatial configuration of converters remains one of the most important factors influencing wave farm efficiency. [9]

While previous studies have focused on optimisation techniques and engineering constraints for wave energy converter arrays, fewer studies have explored predictive modelling of wave farm performance using spatial coordinate data. Therefore, this project investigates whether the spatial positions of wave energy converters can be used to predict the total power output of a wave energy farm using data-driven approaches.

3. Approach Overview

The problem framed as a supervised regression task. The dataset consists of X and Y coordinates, qW (q-factor), individual and total power generated by 49 wave energy converters within a farm. The aim is to predict the farm's total power output on latest and unseen data. This is a practical use case as we will be able to evaluate a spatial layout and estimate the total power generated without physically needing to lay in the WEC's. But because of the high dimensionality of feature space (98 coordinate columns) and non-linear relationship between spatial layout and total power, the model is to be made flexible enough to catch complex patterns.

Primarily, we focused on understanding the data, checking the quality, exploring each variable relationship with the target variable "Total Power" before any modelling decisions were made.

The next phase will proceed as follows:

- **Train/ valuation/ test split:** The cleaned dataset will be divided into training, validation and test dataset probably in an 80/10/10 split fashion to enable the model to pick in most of the complex features. The test data will be held back exclusively for final testing and will not be used during model training or tuning.
- **Baseline model:** A simple dummy regression model will be fitted first to establish a minimum reference point. This will help us assess the performance of a complex model performance to a base model.
- **Model exploration:** A range of regression models will be explored and compared. Models will be selected based on the ability to handle high dimensional inputs as it is necessary to capture the non-linear relationships in our case. We believe Tree based ensemble methods will perform well as it can capture the interactions between multiple spatial features well due to its divide and conquer approach.
- **Hyperparameter tuning:** Once the models will be finalised, we can further tune to optimise performance using cross validation methods and avoid overfitting.
- **Evaluation:** Models will later be assessed using standard regression metrics such as RMSE, R^2, and MAE.
- **Feature importance:** We will examine the contribution of individual spatial features to model's predictions to understand which converter positions matter most, directly addressing research question 3.

4. Data

4.1 Source and Acquisition

The dataset, WEC_Sydney_49.csv, was taken from the UCI Machine Learning Repository [10]. The data is from a 49-converter wave energy farm off the Sydney south coast. The file was uploaded to Google Drive and then loaded to Google Colaboratory notebook using Pandas for analysis. df.shape displayed that the dataset contained 17,964 rows and 149 columns.

4.2 Dataset Structure

All 149 columns were continuous, and of float64 data type.

The columns fall into four major groups:

- **Spatial features (98 columns):** Coordinates for 49 WEC's ranging from X1 to X49 and Y1 to Y49.
- **Individual power outputs (49 columns):** Power1 to Power49 provides data on power generated by individual converter under a given layout.
- **qW:** Efficiency metric, derived from the ration of total power of the farm and the summation of individual power
- **Target variable (Total Power):** The total energy output of the farm in each layout. We will be working on predicting this variable.

4.3 Data Preparation

4.3.1 Missing Values

We observed no null or missing values. No imputation or row removal was required.

4.3.2 Duplicate Removal

We found that 13,148 out of 17,964 were exact duplicates meaning 73.2% of the raw dataset was repeated data. We investigated further by grouping on all columns: some configurations appeared over 100 times, with the most repeated layout occurring 161 times. Keeping the duplicates would cause the model to memorise the most common configurations rather than learn generalised spatial patterns. So, all duplicates were removed, leaving 4,816 unique records.

4.3.3 Feature Selection and Leakage Prevention

Two sets of columns were excluded before modelling:

- **qW:** It is derived directly from power outputs. So, including it would result in data leakage (*Appendix-C*)
- **Power1–Power49:** These are individual outputs and hold no value in predicting the total output of the farm.

The final feature matrix contains only the 98 spatial coordinate columns (X1 to X49, Y1 to Y49), giving a shape of (4,816 × 98). The target vector has 4,816 observations.

4.4 Exploratory Data Analysis

4.4.1 Summary Statistics

We used df.describe() to check on the descriptive statistics and below are the insights.

- The mean and standard deviation of Total Power was observed to be 3.99 MW and 70,705 W, suggesting that precision-level changes are affecting output rather than large swings.
- Individual converter outputs varied much more widely (57,816 W to 110,760 W), suggesting that some positions are consistently outperforming others across different layouts.

4.4.2 Target Variable Distribution (Total_Power)

A histogram of Total Power was plotted with 50 bins each to examine the shape of target distribution. The distribution was observed to be almost bell-shaped with a slight left tail. It is centred around the mean of 3.99 MW. Most configurations produce output within a band of roughly ±150 kW of the mean further confirming that spatial layout drives variation in farm performance.

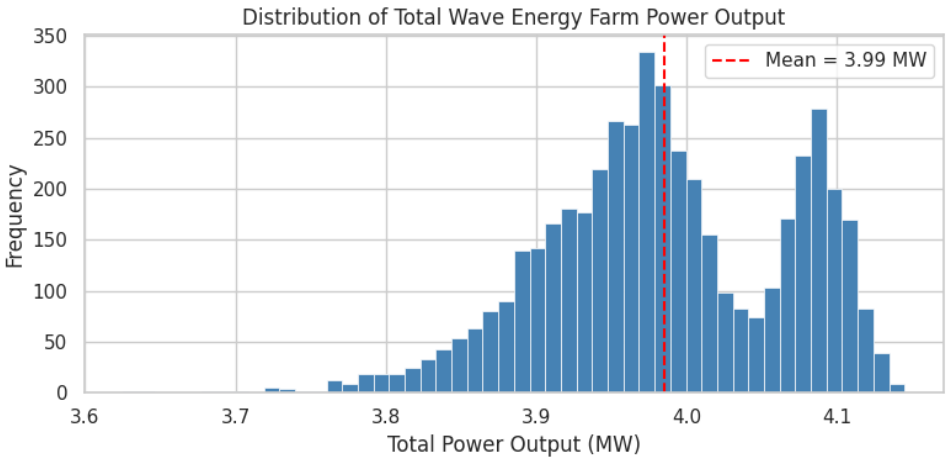

Figure 1: Distribution of Total Power Output

4.4.3 Average Power Output Across Converters

The average power output of each of the 49 converters was calculated across all 4,816 layouts and plotted as a bar chart. (Appendix-A)

Key observations:

- Output varied across converters from below 72,000 W to above 99,000 W on average.
- Some converters consistently outperformed others regardless of the overall layout, suggesting good placement positions.

4.4.4 Spatial Power Heatmap

We melted the farm layouts into long format, to produce 235,984 coordinate power observations (4,816 farms × 49 converters). The coordinate space was divided into a 15×15 grid and mean Total Power for each spatial cell was plotted as a heatmap. Unlike the single-farm scatter plots, this aggregates across the entire dataset, making it the most reliable spatial figure in the analysis.

Key observations:

- Certain regions of the farm boundary are consistently associated with higher total power output.
- The pattern is not uniform, clear hot and cold zones are visible, confirming that converter placement has a structured effect on farm performance.
- This provides strong visual evidence supporting the use of spatial coordinates as predictive features.

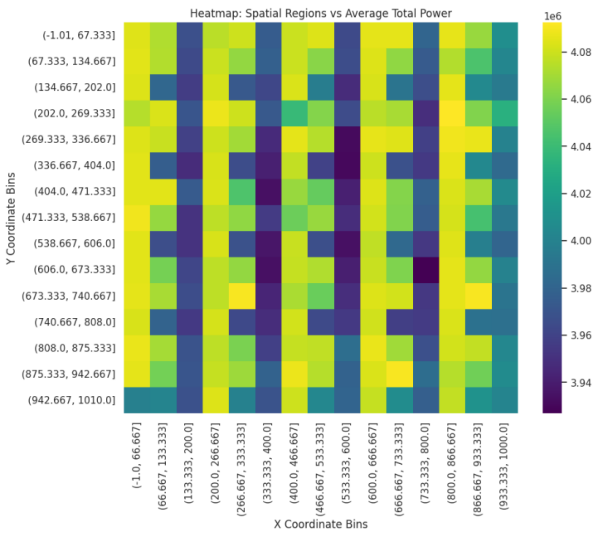

Figure 2: Heatmap: Spatial Regions vs Average Total Power

The hotter zones are seen to cluster towards the edges and corners of the coordinate space, while the central mid-range regions show relatively lower average output. This pattern is consistent with the hydrodynamic interaction effects described in the literature.

4.3.5 Spatial Coordinate Correlations with Total Power

The Pearson correlation between each of the 98 spatial coordinate columns and Total Power was computed. The top positive and top negative correlations were plotted as a bar chart. (*Appendix-B*)

Key observations:

- Both positive and negative correlations were found, hinting that where a converter sits can either help or hurt overall farm output.
- Correlations were modest and spread across many features, consistent with the collective nature of hydrodynamic interactions between converters.
- No single coordinate dominated, supporting the use of the full set of 98 spatial features for modelling.

4.3.6 Single-Farm Spatial Scatter Plots

For the sake of visualising the layout a scatter plot of the first farm was plotted. The first is with all 49 converters coloured by their power output. The second highlights only the top 10 converters.

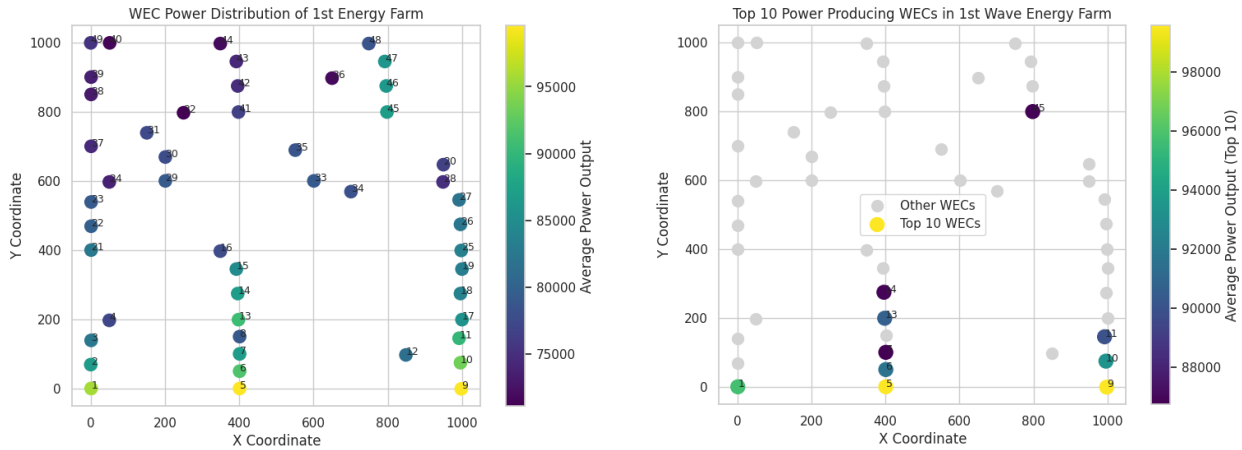

Figure 3: WEC Power Distribution for Farm 1 and Top 10 Power-Producing WECs

4.5 Tools Used

The following libraries were used in Google Colaboratory;

- **Pandas**: For data loading, inspection and feature extraction.
- **NumPy**: For numerical operations supporting the analysis.
- **Matplotlib and Seaborn**: For all visualisations such as histograms, bar charts, scatter plots, and heatmap.

References

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https://doi.org/10.1016/j.rser.2022.112668

[10] UC Irvine Machine Learning Repository - Large-scale Wave Energy Farm
https://archive.ics.uci.edu/dataset/882/large-scale+wave+energy+farm

Appendix

Appendix A - Average Power Output per Converter

Bar chart showing mean power output for each of the 49 WECs across all 4,816 layouts, sorted descending

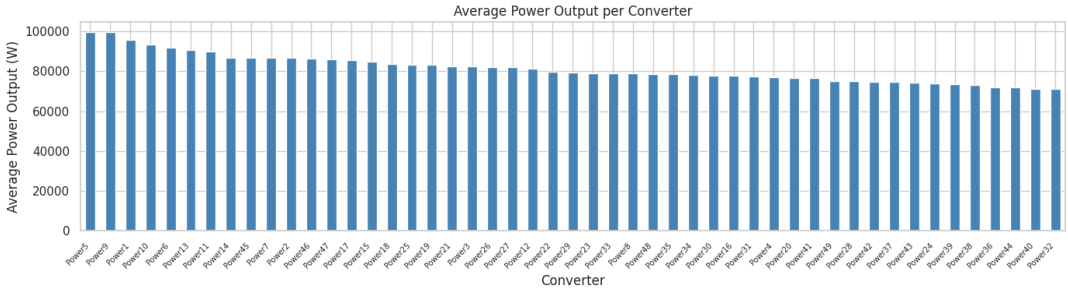

Figure A: Average Power Output per Converter

Appendix B - Top Spatial Coordinates Influencing Total Power

Bar chart of Pearson correlation between each spatial coordinate column and Total_Power, showing the top positive and top negative correlations.

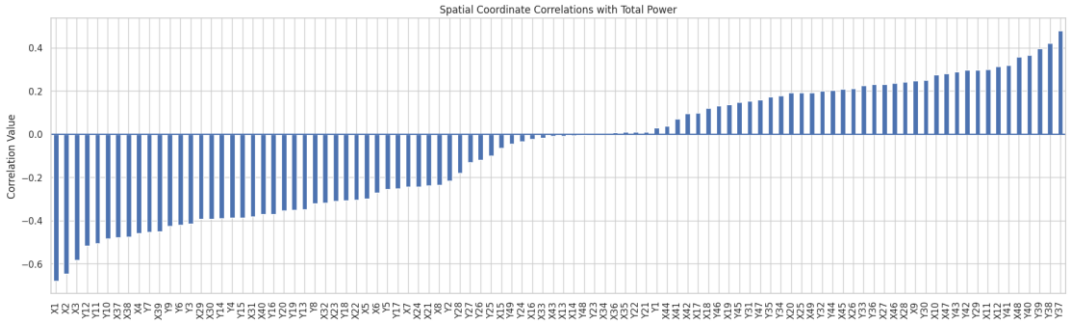

Figure B: Top Spatial Coordinates - Correlation with Total Power

Appendix C - Scatter plot – qW vs Total Power in MW

Scatter plot between qW (q-value) and Total Power in MW shows almost linear relationship between the two.

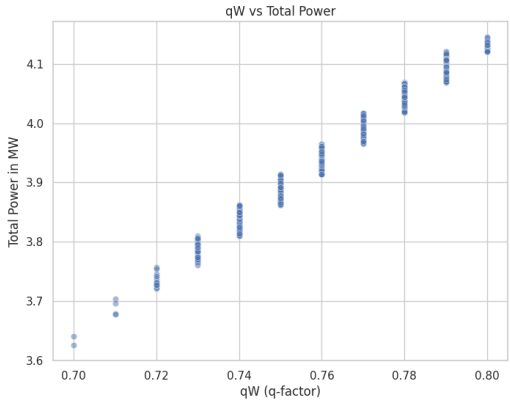

Figure C: Scatter plot – qW vs Total Power in MW

Use of AI Tools

AI tools (e.g., ChatGPT) were used to assist with paraphrasing, improving grammar, and enhancing the presentation of visualisations. All data analysis, code implementation, and interpretations were carried out independently.